

Supplementary Material for MICA: A Model-Informed Change-point Analysis Framework

S1. Quantitative comparison with alternative CPD methods

This section reports the full Turing Change Point Detection (TCPD) benchmark used to compare MICA with the 15 published TCPD competitor algorithms plus an additional HMM-Regime baseline. All 42 datasets, the MICA runner, the extracted competitor tables, and the baseline implementations are archived in `MICA/benchmarking/TCPD/`.

S1.1 Benchmark design

We evaluated MICA on the 42 univariate time series of the TCPD benchmark suite [1]. MICA-O (oracle) selection reports the best F1 over all model, objective, penalty, and hyperparameter configurations per dataset, using the true labels to select the best configuration.

For each dataset, MICA was run with the built-in regression, autoregressive, ETS, and difference-equation models listed in Table S3, using the BIC-like penalty family described in the main manuscript. Baseline numbers for the 15 competitor methods reported in the original TCPD paper (AMOC, BinSeg, BOCPD, BOCPDMS, CPNP, ECP, KernelCPD, PELT, PROPHET, RBOCPDMS, RFPOP, SegNeigh, WBS, and the ZERO baseline) are taken from the extracted per-dataset tables in the original TCPD paper. In addition, we ran one non-TCPD baseline, HMM-Regime, using the official Python package `hmmlearn` (Gaussian hidden Markov model with diagonal covariance; number of states K selected by an oracle grid over $\{2, 3, 4, 5\}$).

Table S1: F1 Score — Oracle Selection

Dataset	AMOC BinSeg			BOCPDMS			CPNP			ECP			KernelCPD			PELT			PROPHET			RBOCPDMS			RFPOP			SegNeigh			WBS			ZERO HMM-Regime			MICA		
bank	1.000	1.000	1.000	0.500	1.000	0.200	0.333	1.000	1.000	0.500	1.000	1.000	0.043	1.000	0.000	1.000	0.578	0.473	0.909	1.000	0.889	0.869	0.857	1.000	1.000	0.889	0.869	0.857	1.000	1.000	1.000	1.000	1.000	1.000	1.000				
bitcoin	0.507	0.780	0.733	0.533	0.611	0.625	0.665	0.783	0.446	0.422	0.727	0.783	0.690	0.450	0.532	0.578	0.473	0.909	1.000	0.889	0.869	0.857	1.000	1.000	0.889	0.869	0.857	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000				
brent_spot	0.465	0.670	0.631	0.309	0.735	0.636	0.553	0.641	0.249	0.621	0.641	0.651	0.564	0.315	0.473	0.473	0.473	0.909	1.000	0.889	0.869	0.857	1.000	1.000	0.889	0.869	0.857	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000				
businv	0.588	0.588	0.650	0.455	0.588	0.370	0.294	0.588	0.275	0.553	0.588	0.588	0.289	0.588	0.214	0.909	1.000	0.889	0.869	0.857	1.000	1.000	0.889	0.869	0.857	1.000	1.000	0.889	0.869	0.857	1.000	1.000	1.000	1.000	1.000				
centralia	0.909	1.000	1.000	1.000	1.000	0.909	1.000	1.000	0.763	0.974	1.000	1.000	0.556	0.763	0.941	1.000	0.556	0.763	0.941	1.000	0.889	0.869	0.857	1.000	1.000	0.889	0.869	0.857	1.000	1.000	1.000	1.000	1.000	1.000	1.000				
children_per_woman	0.678	0.678	0.787	0.826	0.734	0.551	0.525	0.778	0.310	0.580	0.637	0.778	0.500	0.507	0.286	0.889	0.869	0.857	1.000	1.000	0.889	0.869	0.857	1.000	1.000	0.889	0.869	0.857	1.000	1.000	1.000	1.000	1.000	1.000	1.000				
co2_canada	0.544	0.893	0.924	0.479	0.677	0.875	0.867	0.894	0.482	0.621	0.894	0.894	0.681	0.361	0.660	0.869	0.857	1.000	1.000	0.889	0.869	0.857	1.000	1.000	0.889	0.869	0.857	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000				
construction	0.696	0.709	0.709	0.516	0.709	0.709	0.634	0.709	0.324	0.615	0.709	0.709	0.523	0.696	0.706	0.857	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000				
debt_ireland	0.760	1.000	1.000	0.892	0.958	0.980	1.000	1.000	0.469	0.958	1.000	1.000	0.538	0.469	0.947	1.000	0.538	0.469	0.947	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000				
gdp_argentina	0.889	0.947	0.947	0.583	0.889	0.889	0.800	0.947	0.615	0.818	0.947	0.947	0.608	0.652	0.824	0.943	0.608	0.652	0.824	0.943	0.943	0.943	0.943	0.943	0.943	0.943	0.943	0.943	0.943	0.943	0.943	0.943	0.943	0.943	0.943				
gdp_croatia	1.000	0.824	1.000	0.583	1.000	0.824	0.800	0.824	0.824	0.583	0.824	0.824	0.824	0.824	0.824	0.824	0.824	0.824	0.824	0.824	0.824	0.824	0.824	0.824	0.824	0.824	0.824	0.824	0.824	0.824	0.824	0.824	0.824	0.824	0.824				
gdp_iran	0.696	0.696	0.862	0.838	0.667	0.824	0.734	0.808	0.652	0.899	0.833	0.808	0.576	0.652	0.636	1.000	0.576	0.652	0.636	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000				
gdp_japan	1.000	1.000	1.000	0.615	1.000	1.000	0.500	1.000	0.889	1.000	1.000	1.000	0.222	0.889	1.000	0.222	0.889	1.000	0.222	0.889	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000				
global_co2	0.929	0.929	0.929	0.458	0.846	0.929	0.750	0.929	0.463	0.634	0.929	0.929	0.250	0.846	0.800	1.000	0.250	0.846	0.800	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000				
homeruns	0.812	0.889	0.829	0.675	0.681	0.829	0.829	0.829	0.829	0.829	0.829	0.829	0.829	0.829	0.824	0.824	0.829	0.824	0.824	0.824	0.824	0.824	0.824	0.824	0.824	0.824	0.824	0.824	0.824	0.824	0.824	0.824	0.824	0.824	0.824				
iceland_tourism	0.947	0.947	0.947	0.486	0.947	1.000	0.486	0.947	0.220	0.667	0.947	0.947	0.200	0.947	0.000	1.000	0.200	0.947	0.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000				
jfk_passengers	0.776	0.776	0.776	0.650	0.776	0.651	0.651	0.776	0.354	0.420	0.776	0.776	0.437	0.723	0.222	0.857	0.437	0.723	0.222	0.857	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000				
lga_passengers	0.561	0.620	0.715	0.563	0.634	0.892	0.598	0.602	0.366	0.561	0.666	0.620	0.674	0.535	0.684	0.590	0.674	0.535	0.684	0.590	0.590	0.590	0.590	0.590	0.590	0.590	0.590	0.590	0.590	0.590	0.590	0.590	0.590	0.590	0.590				
measles	0.947	0.947	0.947	0.486	0.947	0.080	0.281	0.947	0.391	0.090	0.947	0.947	0.041	0.947	0.071	1.000	0.041	0.947	0.071	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000				
nile	1.000	1.000	1.000	0.800	1.000	1.000	1.000	1.000	0.824	0.800	1.000	1.000	0.824	0.824	0.800	1.000	0.824	0.824	0.800	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000				
ozone	0.776	1.000	0.966	0.778	1.000	1.000	0.667	1.000	0.723	1.000	1.000	1.000	0.286	0.723	0.800	1.000	0.286	0.723	0.800	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000				
rail_lines	0.846	0.966	0.966	0.889	0.966	0.966	0.800	0.846	0.537	0.730	0.889	0.966	0.205	0.537	0.750	1.000	0.205	0.537	0.750	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000				
ratner_stock	0.776	0.832	0.929	0.559	0.776	0.776	0.754	0.824	0.280	0.686	0.824	0.824	0.378	0.571	0.462	0.947	0.378	0.571	0.462	0.947	0.947	0.947	0.947	0.889	0.869	0.857	1.000	1.000	0.889	0.869	0.857	1.000	1.000	1.000	1.000				
robocalls	0.800	0.966	0.966	0.750	0.966	0.966	0.966	0.966	0.636	0.765	0.966	0.966	0.714	0.636	0.857	1.000	0.714	0.636	0.857	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000				
scanline_126007	0.710	0.920	0.964	0.837	0.927	0.870	0.838	0.889	0.644	0.570	0.914	0.940	0.818	0.644	0.804	0.900	0.818	0.644	0.804	0.900	0.900	0.900	0.900	0.900	0.900	0.900	0.900	0.900	0.900	0.900	0.900	0.900	0.900	0.900	0.900				
scanline_42049	0.485	0.931	0.962	0.921	0.809	0.910	0.908	0.910	0.269	0.526	0.910	0.920	0.650	0.276	0.897	0.898	0.650	0.276	0.897	0.898	0.898	0.898	0.898	0.898	0.898	0.898	0.898	0.898	0.898	0.898	0.898	0.898	0.898	0.898	0.898				
seatbelts	0.824	0.838	0.683	0.583	0.838	0.683	0.683	0.683	0.452	0.621	0.683	0.683	0.583	0.621	0.429	0.957	0.583	0.621	0.429	0.957	0.957	0.957	0.957	0															

S1.2 The TCPD benchmark datasets

The Turing Change Point Dataset [1] is a collection of real-world univariate time series assembled for evaluating offline change-point-detection algorithms. Each series was annotated by multiple human experts. The 42 series span several domains:

- **Finance:** bitcoin, ratner_stock;
- **Economics/demographics:** gdp_argentina, children_per_woman, us_population;
- **Environmental monitoring:** co2_canada, global_co2, ozone;
- **Transportation:** jfk_passengers, lga_passengers, rail_lines;
- **Sports and logging:** homeruns, run_log;
- **Quality control:** quality_control_1–quality_control_5.

Following the TCPD protocol, a detected change point is considered correct if it falls within a ± 5 -day window of an annotated change point. Precision is computed against the union of all annotators, while recall is the macro-average over per-annotator ground-truth labels; the reported F1 score balances precision and recall.

S1.3 Mean performance summary

The mean oracle F1 scores across all 42 datasets are summarized in Figure 12 and Table 2 of the main manuscript.

S1.4 Per-dataset F1 scores

Table S1 reports the per-dataset F1 scores under oracle selection. Missing or failed runs in the original TCPD tables are marked with “—” and counted as 0.0 when computing per-method means.

S1.5 Best MICA configuration per dataset

Table S2 lists the model, hyperparameter configuration, and F1 score for the best MICA-O run on each TCPD dataset. The “Config” column is encoded as `ms<min_seg>_s<step>`, i.e. minimum segment length and step size.

S1.6 Qualitative comparison of methods

A qualitative comparison of MICA with representative state-of-the-art CPD methods is provided in the comparison table of the main manuscript.

S1.7 Illustrative TCPD examples

Figure SS1 shows six representative TCPD time series fitted with MICA. The panels were chosen to illustrate several of the model families used in the benchmark (debt-dynamics difference equations, AR(1), and multiplicative Holt–Winters ETS) and a range of change-point patterns. In each panel, the upper subplot shows the standardized observations (gray) and the fitted piecewise model (colored segments), with detected change points marked by dashed red vertical lines. The lower subplot shows the change points annotated by each human expert (black dots, one row per annotator). The panel title reports the dataset name, the model family, the MICA configuration (`ms<min_seg>_s<step>`), the symmetric ± 5 -day F1 score, the number of detected change points (D), and the number of unique annotator change points (T).

Table S2: Best MICA-O (oracle) configuration per TCPD dataset.

Dataset	Model	Config	F1
bank	cubic	ms1_s1	1.000
brent_spot	accelerator	ms2_s5	0.439
businv	ets_mmm	ms1_s1	0.545
centralia	compound_growth	ms1_s1	0.941
children_per_woman	cubic	ms10_s15	0.710
co2_canada	accelerator	ms2_s1	0.827
construction	ets_mmm	ms1_s6	0.857
debt_ireland	compound_growth	ms1_s1	1.000
gdp_argentina	compound_growth	ms5_s1	1.000
gdp_croatia	cubic	ms1_s1	1.000
gdp_iran	ets_mmm	ms1_s1	0.957
gdp_japan	quadratic	ms1_s1	1.000
global_co2	accelerator	ms3_s5	1.000
homeruns	debt_dynamics	ms3_s5	0.796
iceland_tourism	rational	ms3_s3	1.000
jfk_passengers	hill_function	ms2_s23	0.857
lga_passengers	ar1	ms2_s4	0.514
nile	ar3	ms3_s1	1.000
occupancy	accelerator	ms2_s5	0.805
ozone	accelerator	ms2_s1	1.000
quality_control_1	accelerator	ms2_s1	1.000
quality_control_2	ar2	ms2_s1	1.000
quality_control_3	ar1	ms1_s3	1.000
quality_control_4	quadratic	ms2_s5	0.857
quality_control_5	ar3	ms3_s1	1.000
rail_lines	linear	ms5_s1	0.857
ratner_stock	accelerator	ms3_s6	0.947
robocalls	debt_dynamics	ms1_s1	0.857
run_log	ar1	ms7_s7	0.843
scanline_126007	accelerator	ms16_s4	0.820
scanline_42049	accelerator	ms2_s9	0.898
seatbelts	ar1	ms3_s9	0.957
shanghai_license	quadratic	ms1_s1	0.947
uk_coal_employ	accelerator	ms2_s2	0.743
unemployment_nl	accelerator	ms2_s4	0.860
usd_isk	debt_dynamics	ms2_s4	0.909

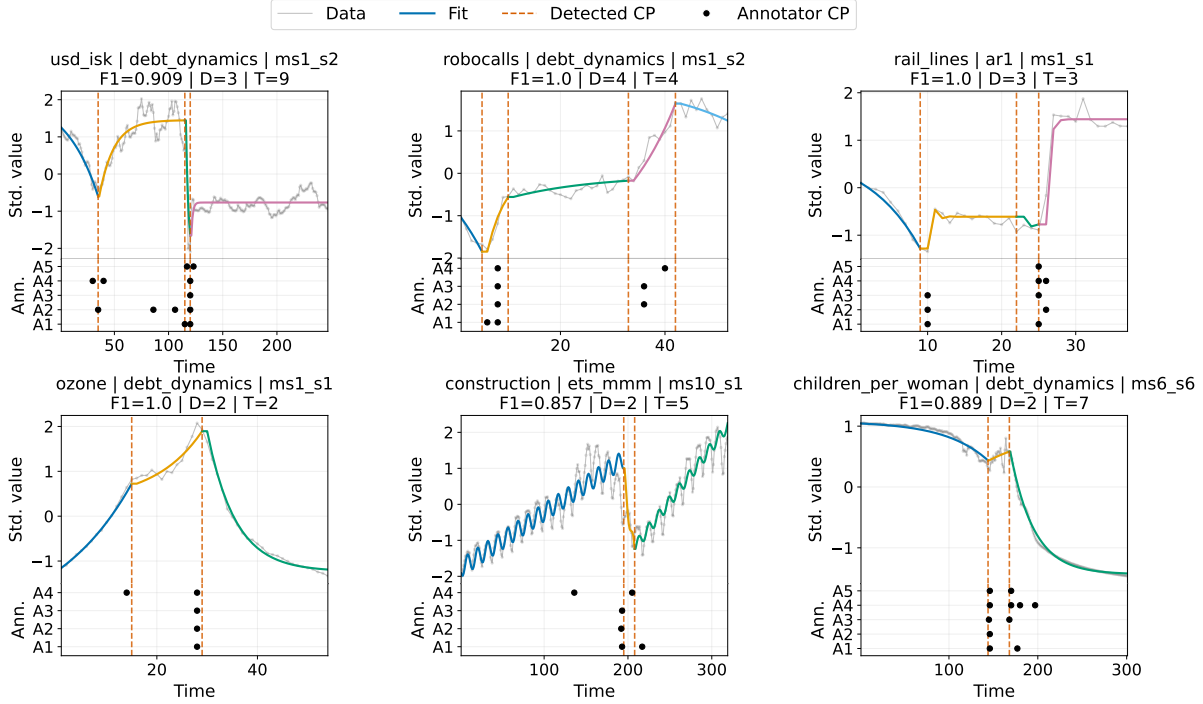


Figure S1: Representative TCPD examples fitted with MICA. Each panel shows the standardized time series (gray), the fitted piecewise model (colored segments), detected change points (dashed red lines), and annotator change points (black dots). The title gives the dataset, model family, configuration, F1 score, number of detected change points (D), and number of unique annotator change points (T).

S2. Built-in segment models in Mica.jl

Table S3 lists the built-in univariate segment models currently available in the `Mica.jl` package. For each model the table gives its model family, a concise equation or description, and the parameter names used by the default setup.

Table S3: Built-in univariate segment models available in `Mica.jl`.

Model	Family	Equation/Description	Parameters
mean	Regression	$y = \mu$	μ
linear	Regression	$y = at + b$	b, a
quadratic	Regression	$y = at^2 + bt + c$	a, b, c
cubic	Regression	$y = at^3 + bt^2 + ct + d$	a, b, c, d
exponential	Regression	$y = c + ae^{bt}$	a, b, c
logistic	Regression	$y = c + K(1 + e^{-r(t-t_0)})^{-1}$	K, r, t_0, c
gompertz	Regression	$y = c + ae^{-be^{-dt}}$	a, b, d, c
saturating exponential	Regression	$y = K(1 - e^{-rt}) + c$	K, r, c
power	Regression	$y = c + at^b$	a, b, c
log linear	Regression	$y = a \log(t) + b$	a, b
mean drift	Regression	$y = \mu + at + b$	μ, a, b
hyperbolic	Regression	$y = a/t + b$	a, b
asymptotic regression	Regression	$y = c + a - be^{-dt}$	a, b, d, c
michaelis menten	Regression	$y = c + V_{\max}t/(K_m + t)$	$V_{\max}, K_m, c$
weibull growth	Regression	$y = c + a - be^{-dt^e}$	a, b, d, e, c
hill function	Regression	$y = c + at^n/(k^n + t^n)$	a, k, n, c

Model	Family	Equation/Description	Parameters
log logistic	Regression	$y = c + a(1 + (t/b)^e)^{-1}$	a, b, e, c
double exponential	Regression	$y = c + ae^{bt} + de^{et}$	a, b, d, e, c
rational	Regression	$y = c + (at + b)/(dt + e)$	a, b, d, e, c
ar1	Autoregressive	$y_t = c + \phi y_{t-1}$	ϕ, c
ar2	Autoregressive	$y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2}$	ϕ_1, ϕ_2, c
ar3	Autoregressive	$y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \phi_3 y_{t-3}$	$\phi_1, \phi_2, \phi_3, c$
exponential ode	ODE	$du/dt = -pu$	p
sir ode	ODE	SIR epidemic ODE: $dS/dt = -\beta SI/N$, $dI/dt = \beta SI/N - \gamma I$, $dR/dt = \gamma I$	β, γ
seir ode	ODE	SEIR epidemic ODE: $dS/dt = -\beta SI/N$, $dE/dt = \beta SI/N - \sigma E$, $dI/dt = \sigma E - \gamma I$, $dR/dt = \gamma I$	β, σ, γ
logistic ode	ODE	$dy/dt = ry(1 - y/K)$	r, K
example difference	Difference	Example difference equation driven by external variables	$\theta_1, \theta_2, \theta_3, \theta_4, \theta_5, \theta_6, \theta_7$
debt dynamics	Difference	$d_t = (1 + r)d_{t-1} - s$	r, s
accelerator	Difference	$y_t = c + v(y_{t-1} - y_{t-2})$	c, v
compound growth	Difference	$y_t = y_{t-1}(1 + r)$	r
garch	Volatility	$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$; $y_t = \mu + \varepsilon_t$	$\mu, \omega, \alpha, \beta$
egarch	Volatility	EGARCH(1,1) log-variance with leverage	$\mu, \omega, \alpha, \beta, \gamma$
tgarch	Volatility	Threshold GARCH(1,1) with asymmetric negative-shock coefficient	$\mu, \omega, \alpha, \beta, \gamma$
poisson	Count	$y_t = c + \exp(a + bt)$	a, b, c
negbin	Count	$y_t = c + \exp(a + bt)$ (negative-binomial mean)	a, b, c
ingarch	Count	$y_t = c + \lambda_t$, $\lambda_t = \omega + \alpha \lambda_{t-1} + \beta \lambda_{t-1}$	ω, α, β, c
ets aaa	ETS	ETS(A,A,A) additive Holt-Winters with deterministic trend and 12-month seasonality	$l_0, b_0, \alpha, \beta, \gamma$
ets mmm	ETS	ETS(M,M,M) multiplicative Holt-Winters with exponential trend and 12-month seasonality	$l_0, b_0, \alpha, \beta, \gamma$
example regression	Regression	$y = at + b$ (example linear model)	a, b

S3. Synthetic toy-dataset benchmark

This section reports the oracle-only comparison of MICA with 15 competitor changepoint-detection algorithms on nine synthetic toy datasets.

S3.1 Toy dataset generation

The nine datasets are generated from three model families, with one seed and three noise levels per family:

- **ODE/SIR** ($n = 163$): SIR model with recovery rate $\gamma = 0.7$, segment-specific transmission rates $\beta \in \{0.00009, 0.00014, 0.00025\}$, true change points at $t = 50$ and $t = 100$, and uniform observation noise at levels 0, 10 and 20.
- **Piecewise-linear regression (LR)** ($n = 201$): continuous piecewise-linear signal with slopes $\{0.8, -0.5, 1.2, -0.8, 0.4\}$, change points at $t = 40, 80, 120, 160$, and Gaussian observation noise at levels 0, 5 and 10.
- **AR(1)** ($n = 300$): AR(1) process with autoregressive coefficients $\{0.0, 0.9, -0.9\}$, change points at $t = 100$ and $t = 200$, and Gaussian observation noise at levels 0, 0.5 and 1.0.

The generators are implemented in `MICA/benchmarking/data/ode_data_generation.jl`, `lr_data_generation.jl`, and `ar_data_generation.jl`. The exported series are stored in `MICA/benchmarking/Toy/data/toy_datasets.json`.

S3.2 MICA settings

MICA was run directly through `Mica.jl` with the 19-objective TCPD-style penalty family described in the main text (`:bic`, `:mdl`, `:aic`, `:tcpd_bic`, `:tcpd_mbic`, `:tcpd_aic`, `:tcpd_hannan_quinn`, `:tcpd_sic`, `:tcpd_none`, `:tcpd_bic0`, `:tcpd_aic0`, `:tcpd_hannan_quinn0`, `:wbs_bic`, `:wbs_mbic`, `:wbs_sic`, `:rfpop_l1`, `:rfpop_l2`, `:rfpop_huber`,

`:rfpop_outlier`). The ODE model used the SIR solver with a genetic-algorithm optimizer. The LR model used the built-in `linear_slope_only` setup with `continuity=true`, so that the fitted piecewise-linear segments share the same continuity architecture as the data generator (each segment has its own slope and the level is propagated across changepoints). The AR model used the built-in `ar1_nodrift` default setup, matching the zero-drift AR(1) process used to generate the AR toy data. For LR and AR, the changepoint grid was searched over the four combinations `min_length/step` = (5, 5), (5, 10), (10, 5) and (10, 10); ODE results used the single grid (10, 10). The oracle result (MICA-O-TCPD) selects, per dataset, the objective and grid with the highest F1 against the true labels.

S3.3 Competitor settings

The 15 competitor algorithms were run through their reference R and Python packages with the oracle hyperparameter grids from van den Burg and Williams [1]:

- R `changepoint` package: AMOC, PELT, BinSeg, SegNeigh.
- R `changepoint.np` package: CPNP.
- R `robseg` package: RFPOP.
- R `ecp` package: ECP and KernelCPD (KCPA).
- R `wbs` package: WBS.
- R `ocp` package: BOCPD.
- Python reference implementations of BOCPDMS and RBOCPDMS from the original authors' GitHub repositories, cloned under `MICA/benchmarking/external/`.
- Python `prophet` package: PROPHET.
- Python `hmmlearn` package: HMM-Regime.
- ZERO baseline: no changepoints.

All competitor methods were evaluated in oracle mode (best configuration per dataset selected by F1). The per-dataset F1 scores are reported in Table S5.

Table S4: Toy datasets: F1 score under oracle selection

Dataset	MICA	PROPHET	RFPOP	PELT	BOCPD	HMM-Regime	CPNP	BOCPDMS	ECP	WBS	KCPA	AMOC	BinSeg	RBOCPDMS	SegNeigh	ZERO
ODE (noise 0.0, $n = 163$)	1.000	0.250	0.222	0.333	0.400	0.200	0.222	0.000	0.182	0.095	0.032	0.000	0.000	0.000	0.000	0.000
ODE (noise 10.0, $n = 163$)	1.000	0.222	0.200	0.333	0.400	0.167	0.125	0.000	0.167	0.095	0.046	0.000	0.000	0.000	0.000	0.000
ODE (noise 20.0, $n = 163$)	1.000	0.250	0.200	0.333	0.333	0.250	0.125	0.000	0.167	0.087	0.078	0.000	0.000	0.000	0.000	0.000
LR (noise 0.0, $n = 201$)	0.667	0.500	0.222	0.235	0.667	0.429	0.273	0.250	0.364	0.276	0.286	0.000	0.222	0.000	0.381	0.000
LR (noise 5.0, $n = 201$)	0.667	0.571	0.375	0.216	0.571	0.286	0.400	0.222	0.044	0.040	0.261	0.000	0.207	0.000	0.333	0.000
LR (noise 10.0, $n = 201$)	0.857	0.500	0.375	0.267	0.571	0.286	0.364	0.333	0.084	0.040	0.058	0.000	0.286	0.000	0.267	0.000
AR (noise 0.0, $n = 300$)	1.000	0.400	0.667	0.667	0.667	0.235	0.364	1.000	0.667	0.444	0.308	0.667	0.667	0.000	0.667	0.000
AR (noise 0.5, $n = 300$)	1.000	0.364	0.500	0.667	0.667	0.308	0.500	1.000	0.571	0.500	0.333	0.667	0.667	0.000	0.667	0.000
AR (noise 1.0, $n = 300$)	0.800	0.308	0.500	0.667	0.667	0.211	0.500	1.000	0.333	0.500	0.200	0.667	0.667	0.000	0.667	0.000
Mean	0.8878	0.3739	0.3623	0.4131	0.5492	0.2634	0.3191	0.4228	0.2865	0.2308	0.1780	0.2222	0.3016	0.0000	0.3312	0.0000

S4. Wind-turbine penalty sensitivity: 15-change-point solution

The wind-turbine analysis in the main manuscript is reported with the BIC/MDL/AIC information criteria, all of which select four change points aligned with strong SCADA status events. Running the same model with the weaker empirical penalty used in the original submission ($\kappa = 100$ in $\text{pen} = \kappa p \log(n)$) produces a denser 15-change-point segmentation. Figure SS2 and Table SS5 reproduce that original result for comparison. Several of the additional change points have weak or no correspondence in the SCADA status logs; they are therefore interpreted as over-segmentation artifacts rather than confirmed physical transitions.

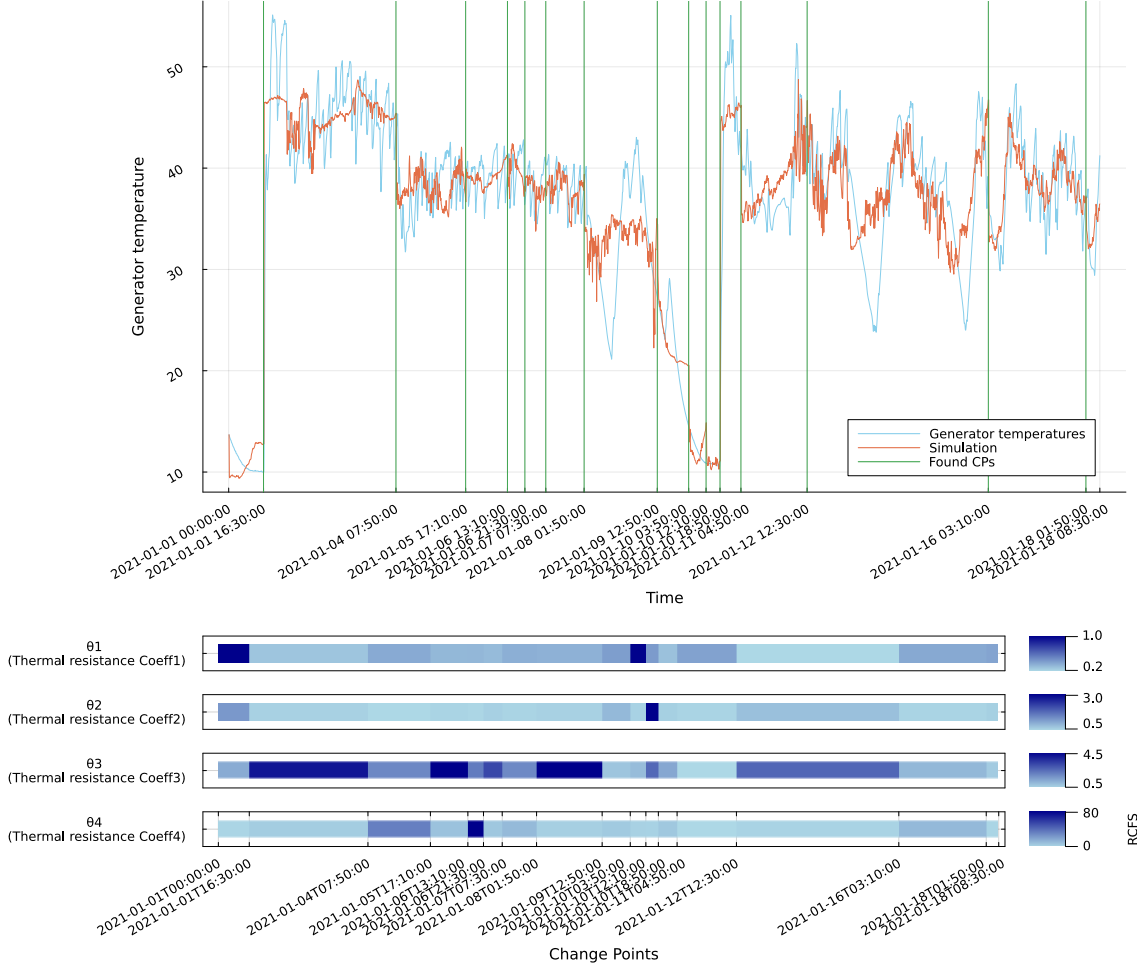


Figure S2: Original 15-change-point wind-turbine segmentation obtained with the weaker empirical penalty ($\kappa = 100$). Top: observed generator temperatures (blue), MICA simulation (orange), and detected change points (green vertical lines). Bottom: relative shifts in segment-specific thermal parameters across segments.

S5. Penalty and information-criterion families in Mica.jl

This section gives the exact formulas for the penalty and information-criterion families supported by `Mica.jl`. All criteria use the least-squares base term $n \log(\hat{\sigma}^2)$ with $\hat{\sigma}^2 = \max\{\text{RSS}/n, 10^{-10}\}$, where RSS is the residual sum of squares returned by the user-supplied loss function. When variance scaling is enabled (the default), the penalty term is multiplied by the null-model variance $\hat{\sigma}_0^2 = \text{RSS}_0/n$, i.e. the residual variance of a constant-mean model on the whole series, so that the objective is comparable across datasets with different scales.

For the built-in criteria `:bic`, `:mdl`, and `:aic`, the total number of free parameters is

$$p_{\text{total}} = n_{\text{global}} + n_{\text{segments}} n_{\text{segment-specific}} + n_{\text{CP}},$$

Change Point	Nearest Status Event	Alignment	Event Type	Description
2021-01-01 16:30	2021-01-01 16:46	Strong	Startup sequence	Change point closely precedes a full startup sequence, indicating a thermal transition as the turbine returns to operation.
2021-01-04 07:50	2021-01-03 22:20	Weak	Data communication unavailable	No direct operational event; data communication issue recorded 9 hours earlier.
2021-01-05 17:10	2021-01-06 23:04	Weak	External stop and restart	Change point occurs more than 24 hours before an external stop/start cycle; not closely aligned.
2021-01-06 13:10	2021-01-06 23:04	Weak	External stop and restart	Occurs 10 hours before a stop/start cycle. Weak alignment.
2021-01-06 21:30	2021-01-06 23:04	Moderate	External stop and restart	Occurs 1.5 hours before turbine stop. Possible lead-in to shutdown.
2021-01-07 07:30	2021-01-07 01:43	Moderate	Frequent external stops	Change point occurs during a dense period of operational cycling; moderate alignment with thermal dynamics.
2021-01-08 01:50	2021-01-08 03:27	Moderate	External stop (low wind)	Change point occurs 1.5 hours before shutdown. Potential early cooling anomaly.
2021-01-09 12:50	2021-01-09 16:26	Weak	Run-up and cycling	Occurs 3.5 hours before operational activity. Not directly aligned.
2021-01-10 03:50	2021-01-10 07:54	Weak	Reset of icing condition	Occurs 4 hours prior to icing status reset. Unclear linkage.
2021-01-10 12:10	2021-01-10 18:48	Weak	Startup after icing	Occurs 6.5 hours before restart. Weak alignment.
2021-01-10 18:50	2021-01-10 18:50	Strong	Gearbox warm-up	Change point aligns exactly with turbine restart and gearbox warm-up sequence.
2021-01-11 04:50	2021-01-10 18:50	Moderate	Extended operation	Follows a long stable run. Possible drift or thermal regime shift.
2021-01-12 12:30	2021-01-12 15:42	Moderate	External stop (low wind)	Change point precedes external stop and restart cycle by 3 hours.
2021-01-16 03:10	2021-01-15 15:30	Weak	Multiple stops	Change point occurs during period of many external stops; no clear triggering event.
2021-01-18 01:50	2021-01-19 12:24	Weak	Yaw and battery test	Occurs more than a day before status activity. No strong alignment.

Table S5: Alignment between the original 15 detected change points (weaker empirical penalty) and turbine status events.

where n_{global} counts non-segment-specific parameters, $n_{\text{segment-specific}}$ counts parameters estimated independently in each segment, $n_{\text{segments}} = n_{\text{CP}} + 1$, and n_{CP} counts the change-point locations.

S5.1 Built-in information criteria

- **BIC:** $n \log(\hat{\sigma}^2) + \hat{\sigma}_0^2 p_{\text{total}} \log(n)$
- **MDL:** $\frac{n}{2} \log(\hat{\sigma}^2) + \hat{\sigma}_0^2 \frac{p_{\text{total}}}{2} \log(n)$
- **AIC:** $n \log(\hat{\sigma}^2) + 2 \hat{\sigma}_0^2 p_{\text{total}}$

With $\hat{\sigma}_0^2 = 1$ (no scaling) these reduce to the standard BIC/MDL/AIC formulas.

S5.2 TCPD-style penalties

These criteria mirror the reference packages used in the Turing Change Point Dataset benchmark. They count only the parameters introduced by a change, not one copy of every segment-specific parameter for every segment. Let d be the number of segment-specific parameters that change at a change point (for MICA this is $n_{\text{segment-specific}}$), $k = n_{\text{CP}}$, and $l_1, \dots, l_{k+1}$ be the segment lengths.

changept / changept.np family (:tcpd_bic, :tcpd_mbic, :tcpd_aic, :tcpd_hannan_quinn, :tcpd_sic, :tcpd_none, and the postfix-0 variants):

- BIC/SIC: $\hat{\sigma}_0^2 (d + 1) \log(n)$
- MBIC: $\hat{\sigma}_0^2 (d + 2) \log(n)$
- AIC: $\hat{\sigma}_0^2 2(d + 1)$
- Hannan-Quinn: $\hat{\sigma}_0^2 2(d + 1) \log(\log(n))$
- BIC0/SIC0: $\hat{\sigma}_0^2 d \log(n)$

- AIC0: $\hat{\sigma}_0^2 2d$
- Hannan–Quinn0: $\hat{\sigma}_0^2 2d \log(\log(n))$
- None: 0

The base likelihood term is the standard least-squares BIC base $n \log(\hat{\sigma}^2)$.

WBS family (:wbs_bic, :wbs_mbic, :wbs_ssic):

- BIC: $\hat{\sigma}_0^2 k \log(n)$
- MBIC: $\hat{\sigma}_0^2 [1.5 k \log(n) + 0.5 \sum_i \log(l_i/n)]$
- SSIC: $\hat{\sigma}_0^2 k (\log(n))^\alpha$, default $\alpha = 1.01$

The WBS base term is $(n/2) \log(\hat{\sigma}^2)$.

RFPOP raw per-change penalty (:rfpop_l1, :rfpop_l2, :rfpop_huber, :rfpop_outlier):

$$\text{objective} = \text{RSS} + \lambda k,$$

where the default λ values follow the TCPD reference implementation: $\lambda = \log(n)$ for L1 and L2, $\lambda = 1.4 \log(n)$ for Huber, and $\lambda = 2 \log(n)$ for the outlier loss.

S5.3 User-defined and legacy penalties

Users may supply an arbitrary penalty function of the segment-specific parameter count p , the data length n , the change-point vector, segment lengths, and/or the number of segments. The legacy empirical penalty used in the original synthetic SIR experiments and in the COVID-19/wind-turbine sensitivity analyses is

$$\text{pen} = \kappa p \log(n),$$

where p was originally the number of segment-specific parameters only. This is retained as an optional sensitivity knob and is not equivalent to the principled BIC/MDL/AIC criteria.

S6. COVID-19 random change-point baseline

To verify that the BIC-selected change points are not an artifact of overfitting, we compared the MICA BIC solution with a random change-point baseline. Keeping the same model, loss, and scaling as the main-text BIC analysis, we drew random change-point sets of size 2–8 and re-optimized the segment-specific parameters. The best 8-change-point random baseline (indices 30, 50, 110, 210, 250, 260, 340, 370) achieved a total loss of 2018.76, more than twice the MICA BIC loss of 870.31 with only two change points. Figure SS3 compares the MICA BIC reference (left) with the best random 8-change-point baseline (right). The random breakpoints fail to capture the two main epidemic waves and produce visibly poorer fits, especially during the first wave and the winter surge. This confirms that the information-criterion-driven segmentation identifies meaningful regime transitions rather than fitting noise.

S7. Identifiability and uncertainty quantification

We assessed parameter identifiability and uncertainty for the COVID-19 Germany model through two complementary analyses: a structural-identifiability study of a rational simplification of the baseline SEIRD model, and a profile-likelihood analysis of the BIC-selected two-change-point solution.

S7.1 Structural identifiability

We used `StructuralIdentifiability.jl` (<https://github.com/SciML/StructuralIdentifiability.jl>) [2], a Julia package implementing the differential-elimination approach of Dong et al. [3] and building on the probabilistic algorithm of Sedoglavic [4] and the theory of identifiability for nonlinear ODE models [5]. It was applied to a polynomial/rational simplification of the 11-compartment SEIRD model used for the COVID-19 analysis. The simplification removes the sinusoidal seasonality term and assumes a constant vaccination rate, because the full model contains non-polynomial (transcendental) terms that are outside the scope of exact structural-identifiability algorithms. All 16 unknown parameters of the simplified model, together with the five observed outputs (cases, hospitalizations, ICU occupancy, deaths, and cumulative vaccinations) and all unobserved state variables, were found to be *globally identifiable* from the assumed observations (Figure SS4). Thus, parameter non-identifiability in the full model is a finite-sample/numerical issue rather than a fundamental structural one.

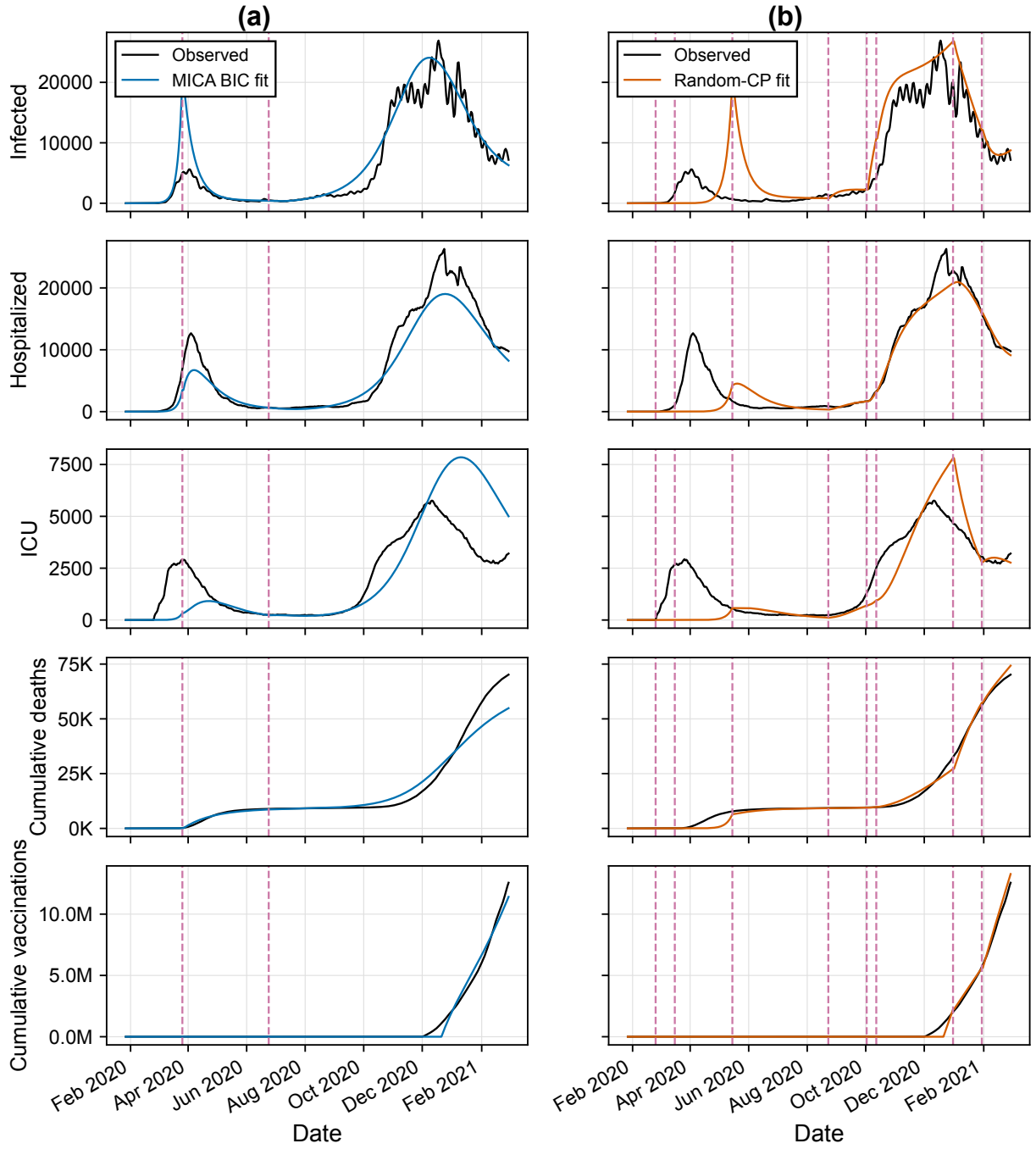


Figure S3: Random change-point baseline for the COVID-19 Germany model. (a) MICA BIC solution with two change points (indices 60 and 150; loss 870.31). (b) Best random 8-change-point baseline (indices 30, 50, 110, 210, 250, 260, 340, 370; loss 2018.76). Both panels show observed data (blue) and the fitted model (orange) on the raw count scale; red dashed vertical lines mark change points.

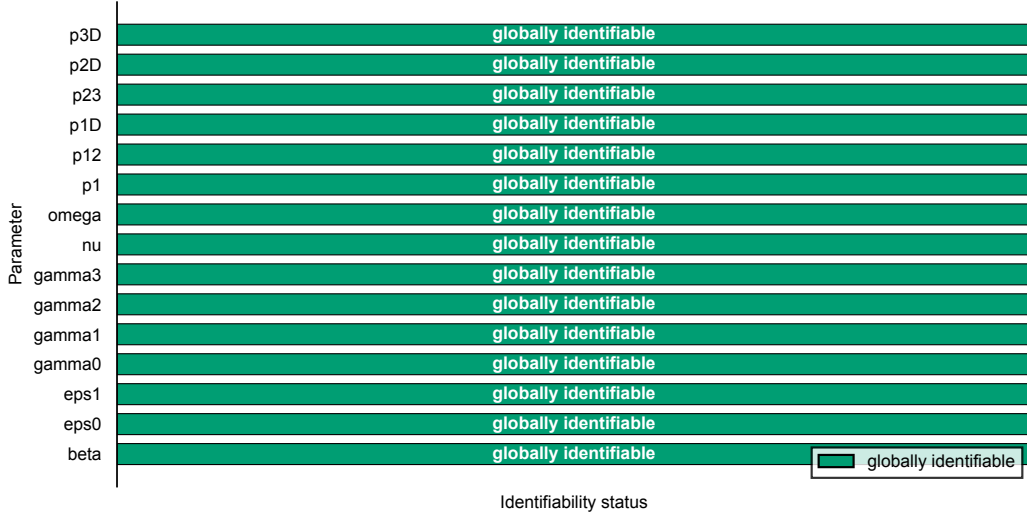


Figure S4: Structural-identifiability assessment of the rational simplification of the baseline COVID-19 SEIRD model (no seasonality, constant vaccination rate). All 16 unknown parameters are globally identifiable from the five assumed observations (cases, hospitalizations, ICU occupancy, deaths, and cumulative vaccinations).

S7.2 Practical identifiability via the Fisher information matrix

A local sensitivity analysis around the best eight-change-point fit (Task E) confirmed that practical identifiability is poor for the full segmented model. The condition number of the Gauss-Newton approximation $J^\top J$ of the Fisher information matrix exceeded 10^{307} , indicating that many parameter combinations produce nearly identical model outputs. The least identifiable directions were dominated by the segment-specific vaccination-rate parameters $\nu_{\text{seg}i}$; several transition probabilities also had very small sensitivity norms. This aligns with the structural result: while every parameter is theoretically identifiable, the available 400-day, five-channel data set does not provide enough information to estimate all 80 free parameters of the richly segmented model reliably.

S7.3 Profile-likelihood confidence intervals

We used `BreakpointProfiles.jl` to compute likelihood-profile confidence intervals for the parameters and change-point locations of the BIC-selected two-change-point solution (change points at indices 60 and 150). The change-point detection was run with an L2 (RMSE) loss and the BIC penalty. The PLE reference fit was then re-optimised under the same L2 loss with a 10-start hybrid GA+LBFGS search; profiles were traced with Nelder-Mead and an adaptive step-out procedure in the style of Data2Dynamics (d2d; <https://github.com/Data2Dynamics/d2d>) [6, 7], using a χ^2 threshold of $\Delta\text{loss} \leq 3.8415$ (95% CI). This gives a consistent Gaussian likelihood interpretation for both detection and uncertainty quantification. The conditional profiles treat the two change points as fixed at their detected values while each parameter is varied; the change-point profiles (Figure SS5) treat each change point as the varied quantity and re-optimize all parameters.

Of the 32 parameters profiled (8 global + 24 segment-specific), 17 were practically identifiable in the conditional profiles: 3 global parameters (ϵ_0 , γ_3 , ω) and 14 segment-specific parameters, including the transmission probabilities $\hat{p}_1$ and scaled transmission rates $\hat{\beta}$ in all three segments. The two change points were both identifiable. Change point 1 (day 60) is pinned tightly to [60, 60]; change point 2 (day 150) has profile interval [146, 150]. The narrowness of the second interval reflects the step = 10 candidate grid used during change-point detection: the continuous likelihood minimum lies near day 148–149, but the reported CP is the nearest grid point at day 150, so the CI is computed relative to that grid-detected reference. The remaining parameters hit either lower or upper bounds, indicating that the data do not constrain them independently given the assumed model structure. This is consistent with the Fisher-information analysis and supports the main-text caveat that detected change points should be interpreted as the best segmentation under the assumed model and criterion, not as independently confirmed structural breaks.

S7.4 True-joint profile-likelihood analysis

As a sensitivity check we also ran a true-joint PLE in which the two change points were treated as free parameters during both the reference re-optimisation and the profiling of every model parameter. The reference fit was improved from an initial loss of 1601.62 to 1594.28; after re-profiling with this improved reference, 12 of the 32 parameters were practically identifiable. The joint change-point profiles remained tight (day 60: [60, 60]; day 150:

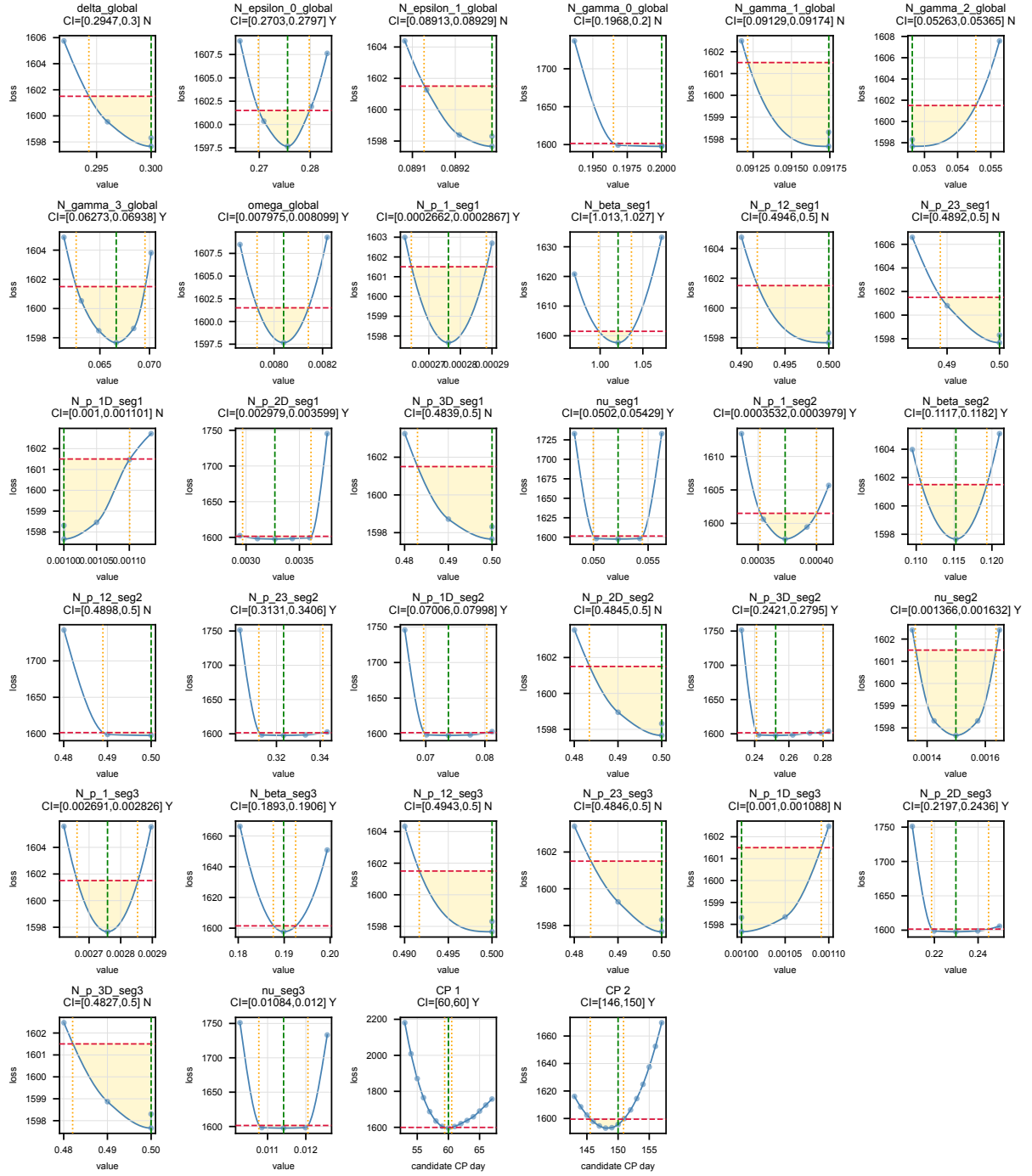


Figure S5: Conditional profile-likelihood curves for all 32 parameters and the two change-point locations of the COVID-19 Germany BIC two-change-point solution. Blue curves are PCHIP-smoothed profile losses; horizontal dashed red lines mark the 95% χ^2_1 threshold; green dashed lines mark the best-fit value; gold shading shows the approximate 95% confidence interval; orange dotted lines show the interval bounds. “Y” and “N” in each panel title indicate whether both bounds were reached inside the inspected range. The change-point panels report intervals relative to the grid-detected reference CPs at days 60 and 150 (change-point search step = 10).

[146, 150]), confirming that the detected transition dates are robust to whether the change points are fixed or co-estimated during uncertainty quantification. Figure SS6 shows the joint PLE curves.

S7.5 Wind-turbine profile-likelihood confidence intervals

We applied the same L2-loss, BIC-selected model and conditional PLE pipeline to the wind-turbine four-change-point solution (change points at indices 140, 500, 1150 and 1860). The reference fit was re-optimised with a 20-start hybrid GA+LBFGS search and profiles were traced with Nelder-Mead, using the same Data2Dynamics-style adaptive step-out and a χ^2_1 threshold of $\Delta\text{loss} \leq 3.8415$ (95% CI). Of the 23 model parameters (4 global + 19 segment-specific), 12 were practically identifiable. All four change points were identifiable: CP 1 [133, 140], CP 2 [493, 501], CP 3 [1150, 1157], CP 4 [1860, 1867]. The profiles are shown in Figure SS7.

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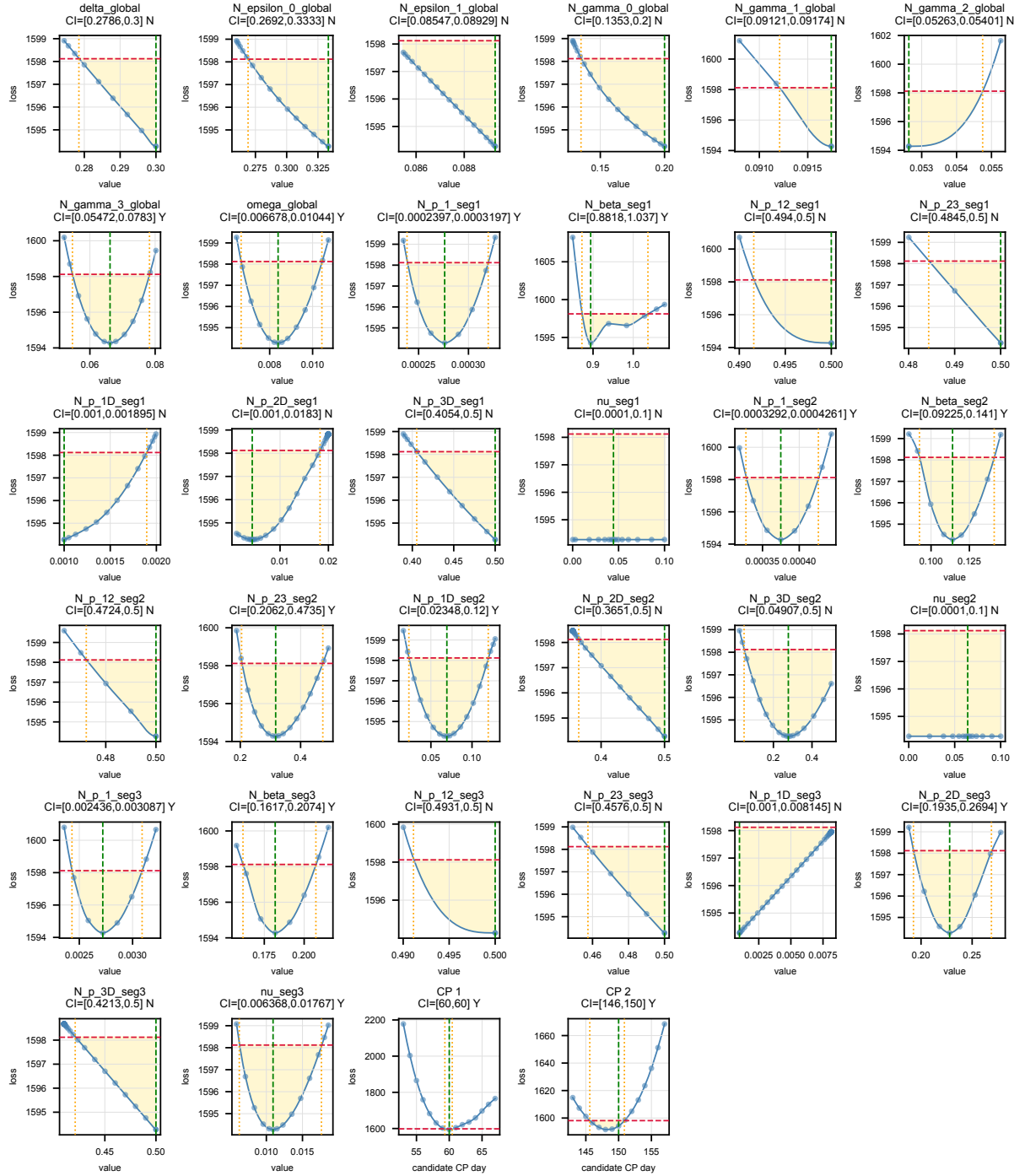


Figure S6: True-joint profile-likelihood curves for all 32 parameters and the two change-point locations of the COVID-19 Germany BIC two-change-point solution. In contrast to the conditional profiles (Figure SS5), the change points are treated as free parameters during both reference optimisation and parameter profiling. Layout and colour coding are as in Figure SS5.

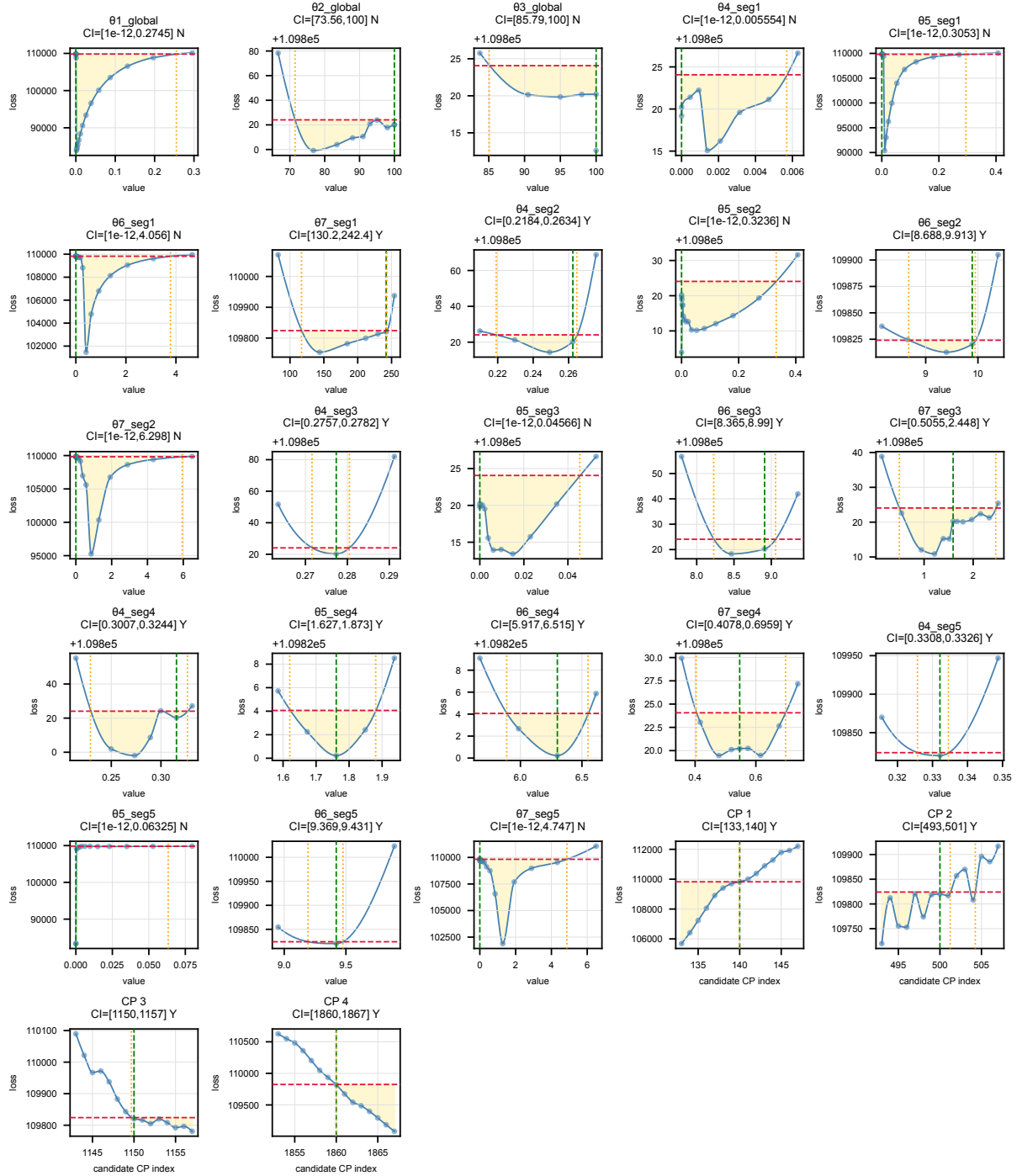


Figure S7: Conditional profile-likelihood curves for all 23 parameters and the four change-point locations of the wind-turbine BIC/MDL/AIC four-change-point solution. Layout and colour coding are as in Figure SS5.